

16. [문제풀이] kube-apiserver 사용 권한 부여하기-2

- 문제

TASK

Find the RBAC permissions needed by the **scraper Deployment** running in the **dev namespace**.

- kubectl logs may help you to find the permissions it needs.

Create a new **ServiceAccount** named **scraper** in the namespace **dev**

- Review the existing **Roles** in the **dev** namespace. and Use best Role to the new ServiceAccount

Update the **scraper Deployment** to use the new **ServiceAccount scraper**

- Deployment에 필요한 RBAC 권한을 찾기
 - Pod의 log를 조회해서, 필요한 권한을 확인
 - 새로운 ServiceAccount를 생성해서, 기존 존재하는 Role들 중, Deployment에 필요한 권한을 식별하여, ServiceAccount에 Role 연결하기
 - Deployment에 새로 만든 ServiceAccount를 적용하기
- ServiceAccount
 - Pod가 k8s API를 호출할 때 사용하는 신원
 - Role
 - 특정 Namespace 안에서 무엇을 할 수 있는지 정의하는 권한 묶음
 - RoleBinding
 - 특정 User/Group/ServiceAccount에게 Role을 연결해 실제 권한을 부여
 - 풀이
 - scraper Deployment에 속한 Pod의 로그 조회
 - default serviceaccount는 pod의 list를 조회할 수 없어서, 에러가 난다.

```
[candidate@k8s-master ~]$ kubectl logs -n dev scraper-56f954bb86-2xcnp
Error from server (Forbidden): pods is forbidden: User "system:serviceaccount:dev:default" cannot list resource "pods" in API group "" in the namespace "dev"
Error from server (Forbidden): pods is forbidden: User "system:serviceaccount:dev:default" cannot list resource "pods" in API group "" in the namespace "dev"
Error from server (Forbidden): pods is forbidden: User "system:serviceaccount:dev:default" cannot list resource "pods" in API group "" in the namespace "dev"
```

◦ scraper Deployment에 속한 Pod 상세 조회

- container의 args를 보면, kubectl get pods 명령어를 실행하는 것을 확인 가능
- spec.serviceAccountName 의 값이 'default' 이다.
- Deployment를 만들 때 Pod의 template 에 serviceAccountName을 미지정하면, Pod의 serviceAccountName 으로 자동으로 'default'가 사용된다.
 - 미지정해서 default로 적용된 경우, Deployment를 상세조회 했을 때는 Pod의 serviceAccountName을 볼 수 없다. Pod를 상세 조회해야 확인 가능.

▼ Pod 상세 조회 결과

```
[candidate@k8s-master ~]$ kubectl get po -n dev scraper-56f954bb86-2xcnp -o yaml
apiVersion: v1
kind: Pod
metadata:
  annotations:
    cnf.projectcalico.org/containerID: 9f069586cc383664171786ec1367db8e6ecbfad649fe71d42e70e35a3511c02c
    cnf.projectcalico.org/podIP: 20.108.82.245/32
    cnf.projectcalico.org/podIPs: 20.108.82.245/32
  creationTimestamp: "2026-08-21T06:37:55Z"
  generateName: scraper-56f954bb86-
  generation: 1
  labels:
    app: scraper
    pod-template-hash: 56f954bb86
  name: scraper-56f954bb86-2xcnp
  namespace: dev
  ownerReferences:
  - apiVersion: apps/v1
    blockOwnerDeletion: true
```

```

    controller: true
    kind: ReplicaSet
    name: scraper-56f954bb86
    uid: 8ea77e99-b973-46e9-8217-9371832a9a15
    resourceVersion: "306741"
    uid: 6064bcf0-6cc9-4b12-ad93-65f764117157
spec:
  containers:
  - args:
    - |
      while true; do
        POD_IPS=$(kubectl get pods -l app=node-exporter -o jsonpath='{.items[*].status.podIP}')
        for ip in $POD_IPS; do
          echo "Call http://$ip:9100/metrics"
          curl -sf "http://$ip:9100/metrics"
        done
        sleep 10
      done
    command:
    - sh
    - -c
    image: bitnami/kubectl
    imagePullPolicy: Always
    name: server
    resources: {}
    terminationMessagePath: /dev/termination-log
    terminationMessagePolicy: File
    volumeMounts:
    - mountPath: /var/run/secrets/kubernetes.io/serviceaccount
      name: kube-api-access-82lms
      readOnly: true
  dnsPolicy: ClusterFirst
  enableServiceLinks: true
  nodeName: k8s-master
  preemptionPolicy: PreemptLowerPriority
  priority: 0
  restartPolicy: Always
  schedulerName: default-scheduler
  securityContext: {}
  serviceAccount: default
  serviceAccountName: default
  terminationGracePeriodSeconds: 30
  tolerations:
  - effect: NoExecute
    key: node.kubernetes.io/not-ready
    operator: Exists
    tolerationSeconds: 300
  - effect: NoExecute
    key: node.kubernetes.io/unreachable
    operator: Exists
    tolerationSeconds: 300
  volumes:
  - name: kube-api-access-82lms
    projected:
      defaultMode: 420

```

```

sources:
  - serviceAccountToken:
      expirationSeconds: 3607
      path: token
  - configMap:
      items:
        - key: ca.crt
          path: ca.crt
          name: kube-root-ca.crt
  - downwardAPI:
      items:
        - fieldRef:
            apiVersion: v1
            fieldPath: metadata.namespace
          path: namespace
status:
  conditions:
    - lastProbeTime: null
      lastTransitionTime: "2026-08-21T06:37:58Z"
      observedGeneration: 1
      status: "True"
      type: PodReadyToStartContainers
    - lastProbeTime: null
      lastTransitionTime: "2026-08-21T06:37:55Z"
      observedGeneration: 1
      status: "True"
      type: Initialized
    - lastProbeTime: null
      lastTransitionTime: "2026-08-21T06:37:58Z"
      observedGeneration: 1
      status: "True"
      type: Ready
    - lastProbeTime: null
      lastTransitionTime: "2026-08-21T06:37:58Z"
      observedGeneration: 1
      status: "True"
      type: ContainersReady
    - lastProbeTime: null
      lastTransitionTime: "2026-08-21T06:37:55Z"
      observedGeneration: 1
      status: "True"
      type: PodScheduled
  containerStatuses:
    - containerID: containerd://2b2f235746385bb8604481486b4f9d9e24b9fc155dd9
      297d9cb84b6316e4ca39
      image: docker.io/bitnami/kubectl:latest
      imageID: docker.io/bitnami/kubectl@sha256:08ec645390af5837b422e5d74db7
      f9b29f072d425baa3cc89c153d0fa87217ab
      lastState: {}
      name: server
      ready: true
      resources: {}
      restartCount: 0
      started: true
      state:
        running:

```

```

    startedAt: "2026-08-21T06:37:58Z"
  volumeMounts:
    - mountPath: /var/run/secrets/kubernetes.io/serviceaccount
      name: kube-api-access-82lms
      readOnly: true
      recursiveReadOnly: Disabled
  hostIP: 192.168.56.41
  hostIPs:
    - ip: 192.168.56.41
  observedGeneration: 1
  phase: Running
  podIP: 20.108.82.245
  podIPs:
    - ip: 20.108.82.245
  qosClass: BestEffort
  startTime: "2026-08-21T06:37:55Z"

```

- 해당 namespace에 기존 존재하는 role들을 조회
 - pod의 list 권한이 있는, 'pod-list' role이 필요하다.

▼ 조회 결과

```

[candidate@k8s-master ~]$ kubectl get role -n dev -o yaml
apiVersion: v1
items:
- apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    creationTimestamp: "2026-08-21T06:37:57Z"
    name: event-reader
    namespace: dev
    resourceVersion: "306734"
    uid: d5d4fe2a-b0e2-4570-b0b8-6a9f063b9ccd
  rules:
  - apiGroups:
    - ""
    resources:
    - events
    verbs:
    - get
    - list
- apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    creationTimestamp: "2026-08-21T06:37:58Z"
    name: pod-list
    namespace: dev
    resourceVersion: "306738"
    uid: 7ad7ef79-d54f-42a6-8110-075331be13eb
  rules:

```

```

- apiGroups:
  - ""
  resources:
  - pods
  verbs:
  - get
  - list
- apiVersion: rbac.authorization.k8s.io/v1
  kind: Role
  metadata:
    creationTimestamp: "2026-08-21T06:37:58Z"
    name: pod-log
    namespace: dev
    resourceVersion: "306740"
    uid: 3dc57df0-cd56-4ee4-af87-ffb48274160b
  rules:
  - apiGroups:
    - ""
    resources:
    - pod/log
    verbs:
    - get
  kind: List
  metadata:
    resourceVersion: ""

```

◦ serviceAccount 생성

- 'default' serviceAccount가 기본적으로 있기 때문에, 해당 serviceAccount의 yaml을 참고해서 작성할 수 있다.

```

apiVersion: v1
kind: ServiceAccount
metadata:
  name: scraper
  namespace: dev

```

◦ RoleBinding 생성

- k8s 문서에 'rbac' 으로 검색해서 나오는 문서를 참조하여 작성한다.
<https://kubernetes.io/docs/reference/access-authn-authz/rbac/>

```

apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding

```

```

metadata:
  name: scraper-rolebinding
  namespace: dev
subjects:
- kind: ServiceAccount
  name: scraper
  namespace: dev
roleRef:
  kind: Role
  name: pod-list
  apiGroup: rbac.authorization.k8s.io

```

- Deployment에 새로 만든 ServiceAccount를 적용

```
kubectl edit deploy -n dev scraper
```

- ▼ spec.template.spec.serviceAccountName 에 'scraper' 로 작성한다.

```

apiVersion: apps/v1
kind: Deployment
metadata:
  annotations:
    deployment.kubernetes.io/revision: "2"
  creationTimestamp: "2026-08-21T06:37:55Z"
  generation: 2
  name: scraper
  namespace: dev
  resourceVersion: "314407"
  uid: 962c4fc7-48ef-4098-9c8e-f27ab3aa28c5
spec:
  progressDeadlineSeconds: 600
  replicas: 1
  revisionHistoryLimit: 10
  selector:
    matchLabels:
      app: scraper
  strategy:
    rollingUpdate:
      maxSurge: 25%
      maxUnavailable: 25%
    type: RollingUpdate
  template:
    metadata:
      labels:
        app: scraper
    spec:
      containers:
      - args:
        - |
          while true; do
            POD_IPS=$(kubectl get pods -l app=node-exporter -o jsonpath='{.

```

```

items[*].status.podIP}')
    for ip in $POD_IPS; do
        echo "Call http://$ip:9100/metrics"
        curl -sf "http://$ip:9100/metrics"
    done
    sleep 10
done
command:
- sh
- -C
image: bitnami/kubectl
imagePullPolicy: Always
name: server
resources: {}
terminationMessagePath: /dev/termination-log
terminationMessagePolicy: File
dnsPolicy: ClusterFirst
restartPolicy: Always
schedulerName: default-scheduler
securityContext: {}
serviceAccount: scraper
serviceAccountName: scraper
terminationGracePeriodSeconds: 30
status:
  availableReplicas: 1
  conditions:
  - lastTransitionTime: "2026-08-21T06:37:58Z"
    lastUpdateTime: "2026-08-21T06:37:58Z"
    message: Deployment has minimum availability.
    reason: MinimumReplicasAvailable
    status: "True"
    type: Available
  - lastTransitionTime: "2026-08-21T06:37:55Z"
    lastUpdateTime: "2026-08-21T07:47:50Z"
    message: ReplicaSet "scraper-6975d5d688" has successfully progressed.
    reason: NewReplicaSetAvailable
    status: "True"
    type: Progressing
  observedGeneration: 2
  readyReplicas: 1
  replicas: 1
  terminatingReplicas: 0
  updatedReplicas: 1

```

- Deployment 변경 후 pod의 log를 조회 해보면, 권한이 추가되어서, 에러가 나지 않음

```

[candidate@k8s-master ~]$ k logs -n dev scraper-6975d5d688-dhch9
process_max_fds 1.073741816e+09
# HELP process_network_receive_bytes_total Number of bytes received by the process over the network.
# TYPE process_network_receive_bytes_total counter
process_network_receive_bytes_total 4.43675708e+08

```



```
# HELP process_network_transmit_bytes_total Number of bytes sent by the process over the network.
# TYPE process_network_transmit_bytes_total counter
process_network_transmit_bytes_total 3.10409986e+08
# HELP process_open_fds Number of open file descriptors.
# TYPE process_open_fds gauge
process_open_fds 8
```